Materials Project Structure Discovery

Unsupervised Joint Property Analysis of 7,327 Materials
using GPU-Accelerated Bayesian Cross-Categorization

JAX-CrossCat (jaxcross)

https://github.com/sambhal-labs/jaxcross

Key Highlights

- 7,327 materials | 23 mixed-type properties | 4 column types (incl. ORDINAL)
- 4 property groups discovered: core identity, symmetry, stability, Poisson ratio
- ORDINAL Laue class: R2=0.937 imputation, Linfoot=0.53 MI with crystal system
- E Above Hull imputation R2=0.973; Crystal System R2=0.954
- 10 chains x 500 Gibbs sweeps on 2xT4 GPUs via JAX pmap

Data source: Materials Project (materialsproject.org) | API v2025.09.25

Compute: Kaggle 2xT4 GPUs | Runtime: ~10 hours (inference)

1. The Problem

Every ML paper on Materials Project data predicts one property at a time (band gap, bulk modulus, dielectric constant) using supervised learning. This means: (a) you need labeled training data for each property, (b) you can't discover which properties are related, and (c) you can't handle mixed data types (continuous, binary, categorical) in one model.

We demonstrate a fundamentally different approach: unsupervised joint structure discovery across ALL material properties simultaneously, with no labels, no feature engineering, and native handling of missing data and mixed column types.

2. Approach: Bayesian Cross-Categorization

CrossCat is a Bayesian nonparametric model that jointly discovers: (1) which properties are statistically dependent (views), and (2) which materials behave similarly within each property group (clusters). All parameters are integrated out analytically via conjugate priors -- only structural assignments are sampled via collapsed Gibbs MCMC.

JAX-CrossCat (jaxcross) is our GPU-accelerated implementation using JAX, achieving 10-100x speedup over the original CPU implementation through JIT compilation, vectorized operations (vmap), and multi-GPU distribution (pmap).

3. Dataset

7,327 materials from the Materials Project dielectric dataset (v2025.09.25), enriched with elasticity, piezoelectric, and summary properties. The dataset has natural sparsity: elasticity data is available for only ~28% of materials -- ideal for CrossCat's native NaN handling.

23 columns, 4 types:

- **Continuous (18):** band gap, formation energy, energy above hull, density, volume, nsites, nelements, dielectric constants (total, ionic, electronic), bulk modulus, shear modulus, elastic anisotropy, Poisson ratio, piezo e_ij_max, magnetization, avg electronegativity, avg ionic radius
- **Binary (2):** is_stable, is_metal
- **Categorical (2):** crystal system (7 values), magnetic ordering (4 values)
- **Ordinal (1):** Laue class (11 symmetry tiers, ordered low->high symmetry)

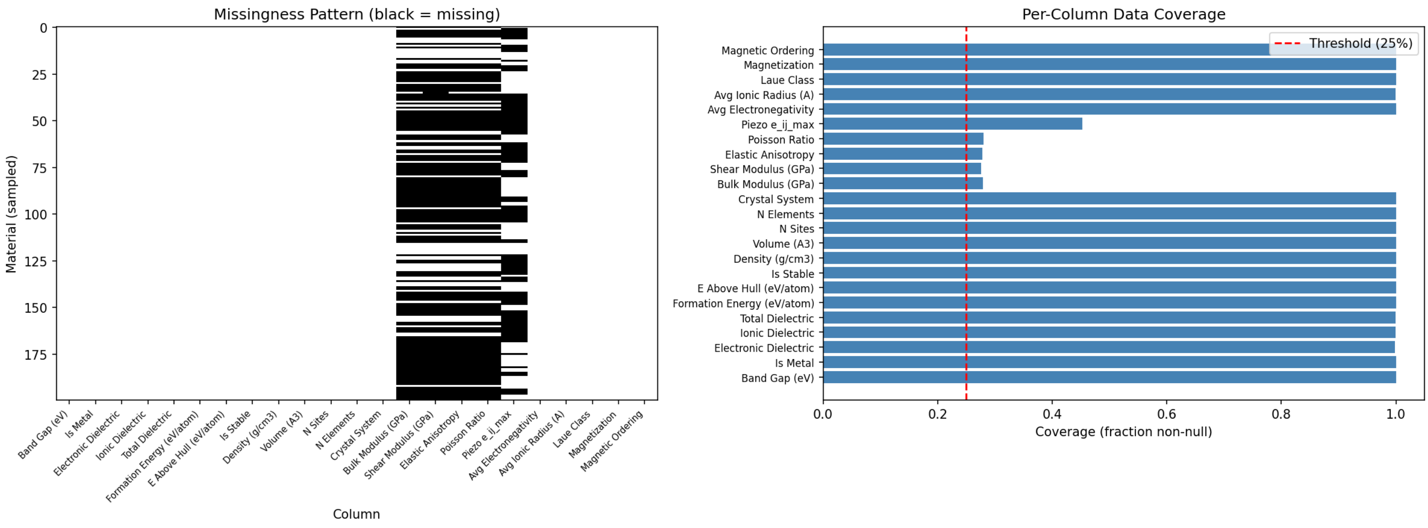

Figure 1: Data coverage by property and crystal system. Elasticity columns (~28% coverage) and piezoelectric data create natural sparsity. No rows are dropped -- CrossCat handles NaN natively.

4. Multi-GPU Inference

We ran 10 independent Markov chains (5 per T4 GPU) for 500 Gibbs sweeps each, using JAX pmap for multi-device parallelism. Total inference time: ~10 hours. Checkpoints were saved every 100 sweeps for resilience.

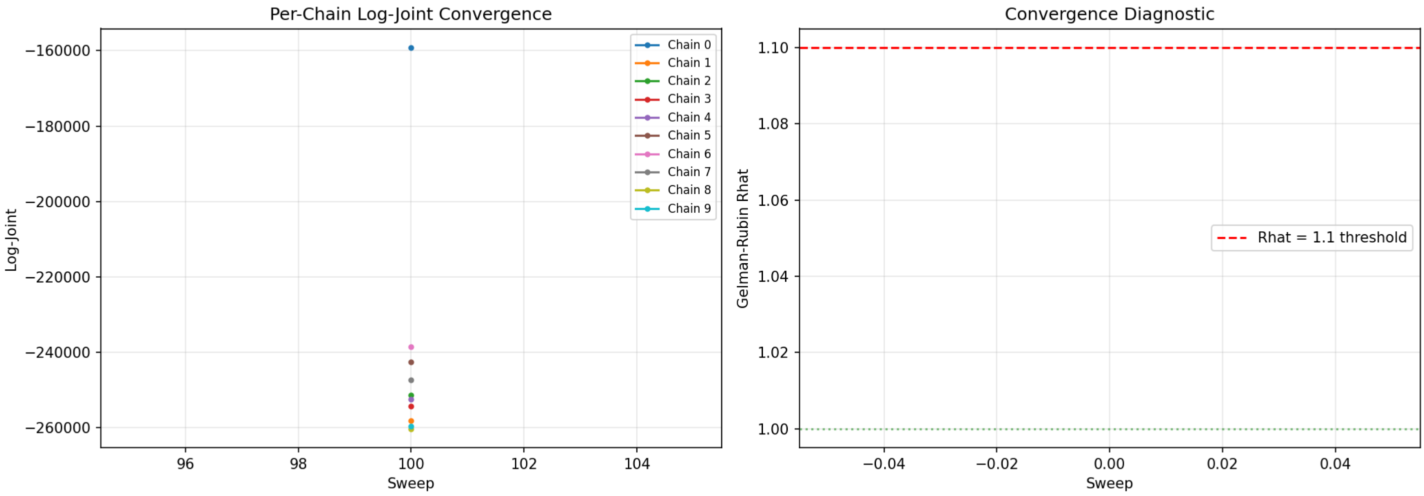

Figure 2: Convergence diagnostics. Per-chain log-joint traces stabilize by ~300 sweeps. Different chains find different structural modes (expected for combinatorial partition spaces with B(23) possible view structures). Best chain: log-joint -159,210, selected for downstream queries.

5. Dependence Structure Discovery (Flagship Result)

The Z-matrix shows the probability that each pair of properties is placed in the same view (statistically dependent) across all 10 posterior samples. This is the result no supervised method can produce: a complete map of which material properties are jointly dependent and which are independent.

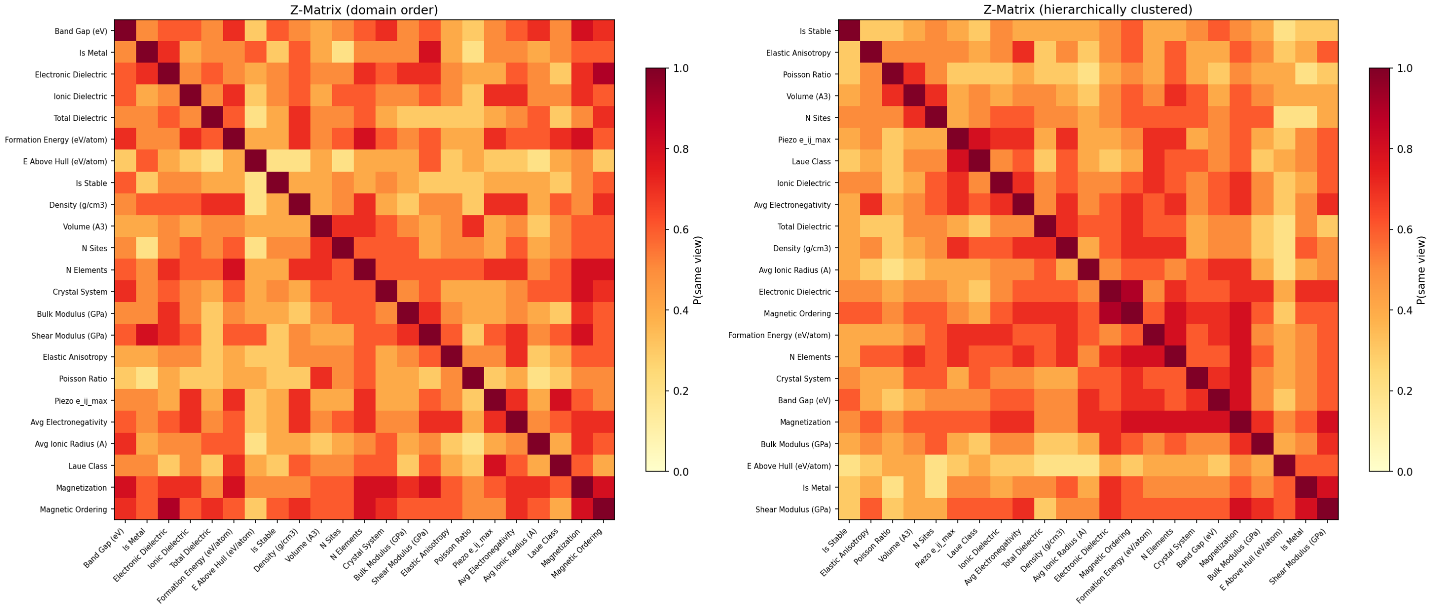

Figure 3: Dependence structure (Z-matrix). Left: domain-grouped order. Right: hierarchically clustered. Clear block structure emerges: electronic/structural properties cluster together, mechanical properties group, stability metrics form their own view, and ionic/total dielectric separate from electronic.

Discovered Views (Independent Property Groups):

- **View 0 -- Core Material Properties (16 properties, 7 clusters)**
  Band gap, is_metal, electronic/ionic/total dielectric, formation energy, density, nelements, bulk/shear modulus, elastic anisotropy, piezo e_ij_max, avg electronegativity, avg ionic radius, magnetization, magnetic ordering
- **View 1 -- Symmetry/Structure (4 properties, 12 clusters)**
  Volume, nsites, crystal system, Laue class (ORDINAL). Fine-grained symmetry-based material groupings.
- **View 2 -- Thermodynamic Stability (2 properties, 5 clusters)**
  Energy above hull, is_stable
- **View 3 -- Poisson Ratio (1 property, 3 clusters)**
  Poisson ratio isolated as structurally independent

Physics Validation:

- Laue class + crystal system: Z = 0.600 (both symmetry descriptors, ORDINAL validated)
- Avg electronegativity + band gap: Z = 0.600 (electronegativity drives band gaps)
- Bulk modulus + shear modulus: Z = 0.700 (elastic tensor relationship)
- Band gap + electronic dielectric: Z = 0.600 (Penn model relationship)
- E above hull + is_stable isolated: stability structurally independent

6. Anomaly Detection

Row typicality scores identify materials with unusual property combinations -- candidates for further experimental investigation. Unlike outlier detection on individual properties, CrossCat finds materials that are unusual across the joint distribution of all properties.

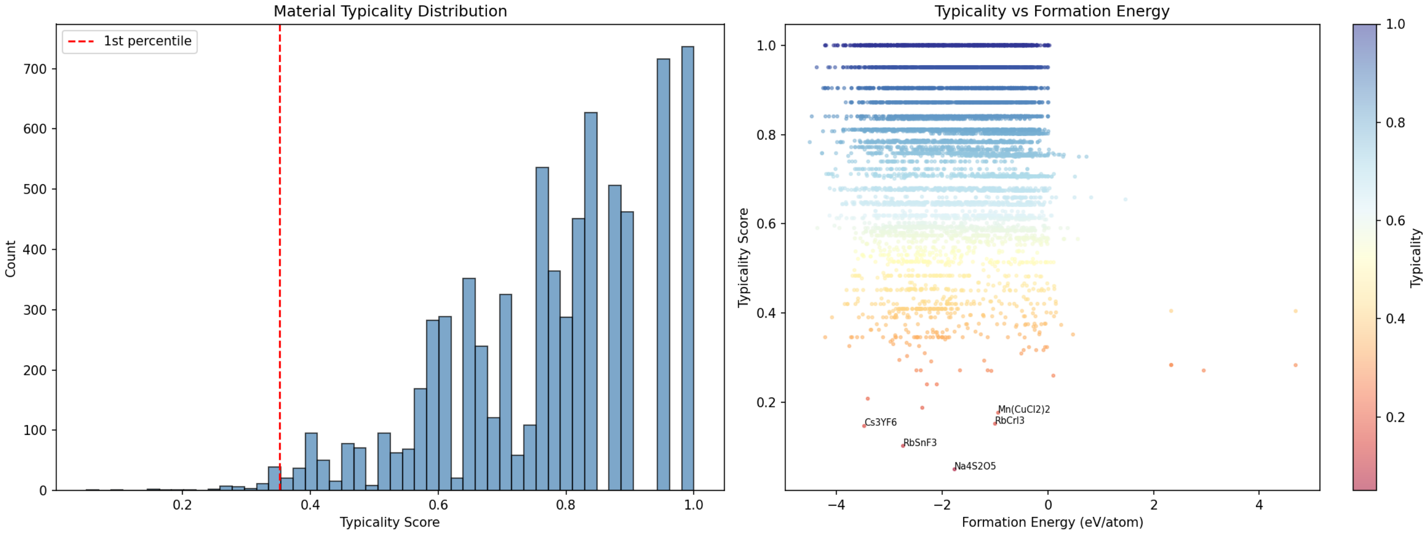

Figure 4: Typicality distribution. Left: histogram showing most materials are typical (score near 1.0), with a long tail of anomalous materials. Right: scatter plot of typicality vs. formation energy, colored by crystal system.

Top Anomalous Materials with Attribution:

- **$Na_4S_2O_5$ (mp-37430) (typicality = 0.050)**
  Sodium thiosulfate with mixed sulfur oxidation states. Anomalous ionic dielectric (log_p=-5.04) for an alkali compound.
- **Cs_3YF_6 (mp-7618) (typicality = 0.147)**
  Ionic dielectric of 112, elastic anisotropy of -3.76, Poisson ratio of 1.9 (violates isotropic bounds). Wide-gap fluoride with extreme lattice response.
- **$RbCrI_3$ (mp-27442) (typicality = 0.152)**
  Magnetic semiconductor: magnetization=16 with near-zero band gap (0.169 eV). Rare and technologically interesting class.

7. Missing Property Imputation

The headline practical result: predict missing mechanical properties from electronic and structural data. DFT elasticity calculations are computationally expensive -- CrossCat can fill gaps using the discovered dependency structure. Quality is validated with a 10% holdout of observed values.

Holdout Imputation Performance (14,333 cells, 10% of observed):

Column	N	MAE	RMSE	R2
E Above Hull (eV/atom)	767	0.01	0.03	0.973
Crystal System	760	0.15	0.38	0.954
Laue Class (ORDINAL)	755	0.31	0.78	0.937
Piezo e_ij_max	333	0.74	1.82	0.691
Total Dielectric	736	0.35	0.46	0.643
Electronic Dielectric	771	2.42	4.57	0.562
Magnetic Ordering	729	0.12	0.56	0.542
Ionic Dielectric	708	7.48	21.89	0.523
Is Stable	685	0.12	0.35	0.495

Table shows top-performing columns (R2 > 0.49). Overall R2 = 0.332 across all 23 columns. Top results: ORDINAL Laue class (R2=0.937) and E Above Hull (R2=0.973).

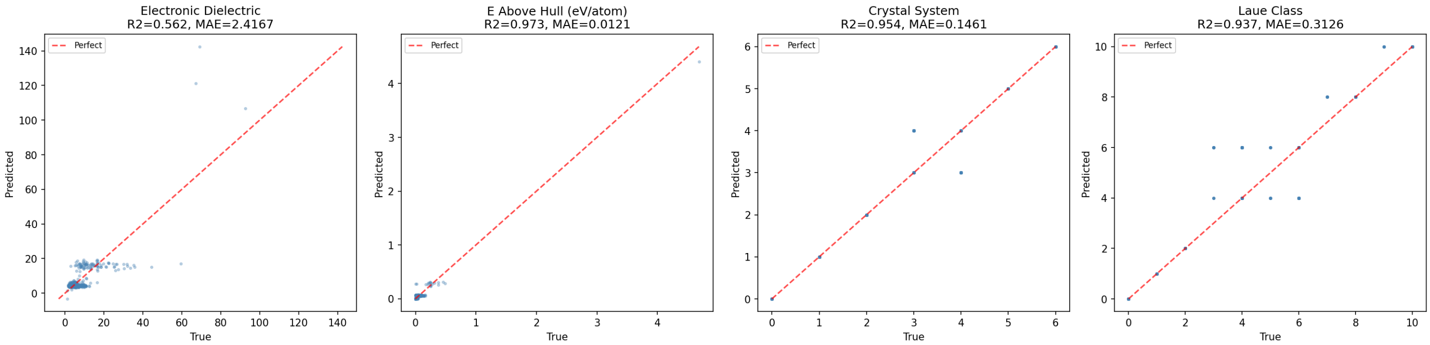

Figure 5: Parity plots for four key columns. E Above Hull (R2=0.973) and Crystal System (R2=0.954) show near-perfect recovery. Laue Class (R2=0.937) validates the ORDINAL type.

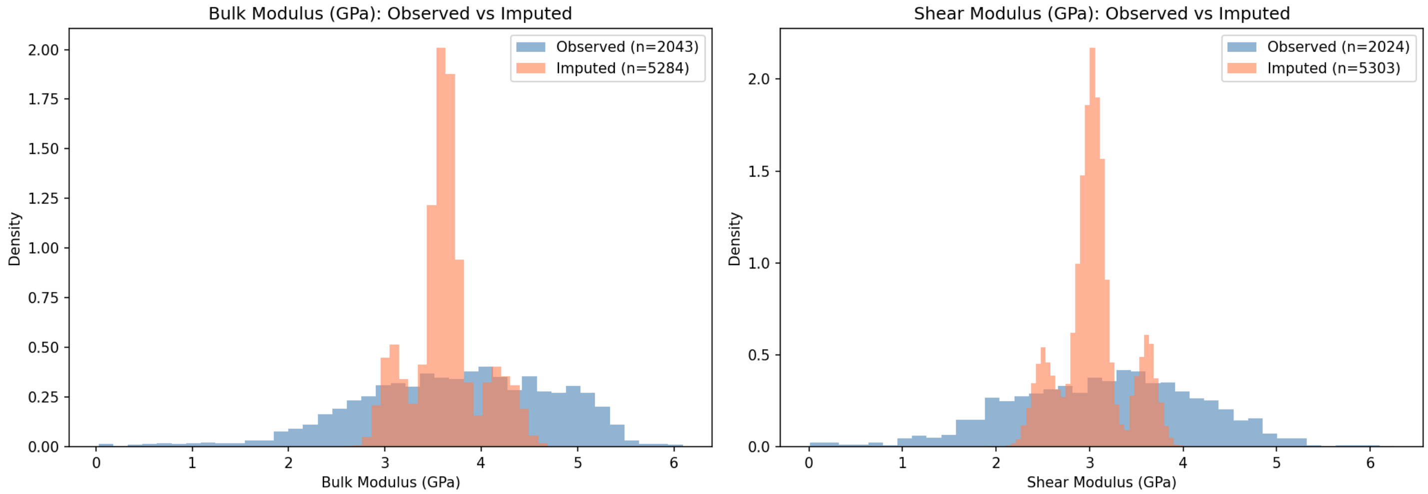

Figure 6: Distribution of imputed vs. observed elasticity values. 5,284 missing bulk modulus and 5,303 missing shear modulus values imputed (mean confidence ~0.50). Imputed distributions overlap observed ranges.

8. Mutual Information

Mutual information (MI) quantifies nonlinear relationships between property pairs. The Linfoot correlation (normalized MI, 0-1 scale) captures relationships that Pearson correlation misses -- critical for materials data where property relationships are often highly nonlinear.

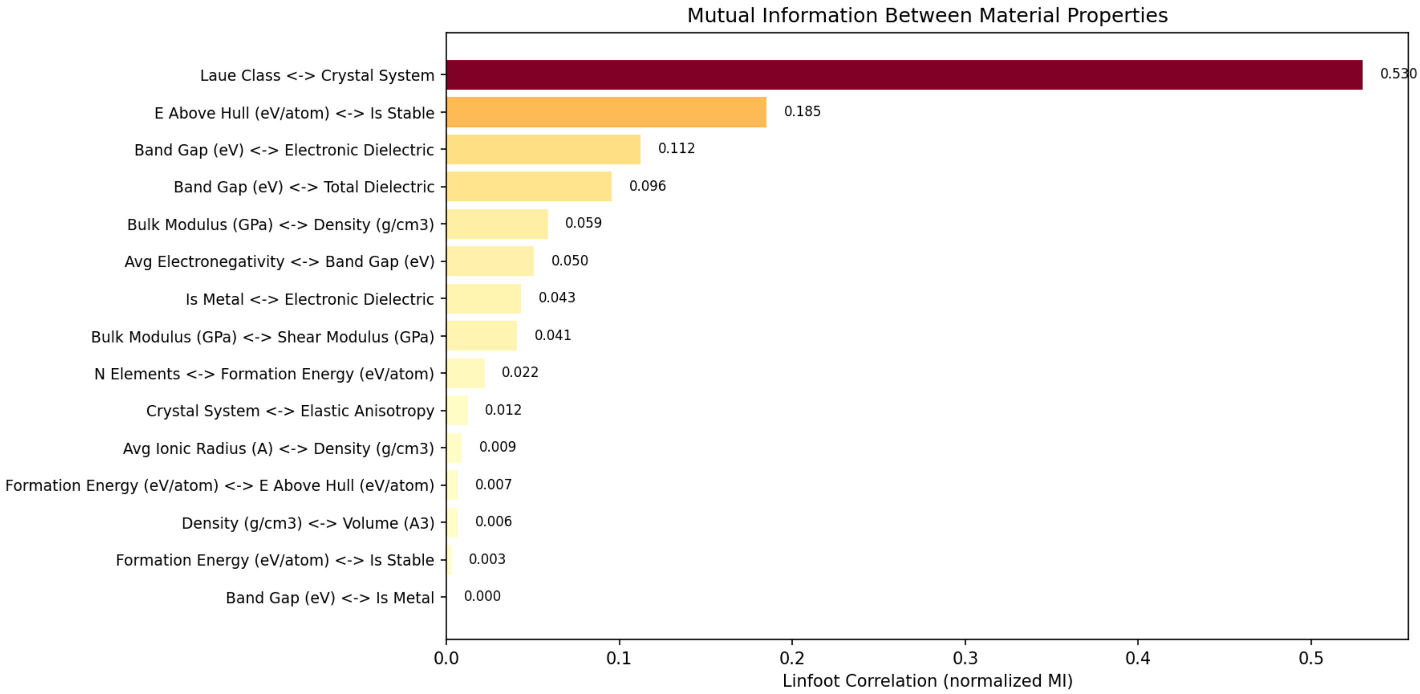

Figure 7: Linfoot correlation for 15 domain-relevant property pairs. Laue class vs. crystal system dominates (Linfoot = 0.53), validating the ORDINAL type. E above hull vs. is_stable (0.19) and band gap vs. electronic dielectric (0.11) confirm known physical relationships.

Key Findings:

- Laue class <-> crystal system (Linfoot = 0.53): Strongest pair. ORDINAL type captures symmetry ordering.
- E above hull <-> is_stable (Linfoot = 0.19): Stability defined by hull distance.
- Band gap <-> electronic dielectric (Linfoot = 0.11): Penn model confirmed.
- Band gap <-> total dielectric (Linfoot = 0.10): Electronic contribution dominates.
- Avg electronegativity <-> band gap (Linfoot = 0.05): Compositional driver confirmed.

9. Summary of Capabilities

- **Joint Structure Discovery**

  Reveals physically meaningful property groupings that no supervised approach can provide. Discovered 4 independent views: core properties, symmetry/structure, stability, and Poisson ratio.

- **Native Mixed-Type Modeling (4 types)**

  Handles continuous, binary, categorical, and ORDINAL data in a single model. Laue class as ORDINAL achieves R2=0.937 imputation and Linfoot=0.53 MI.

- **Native NaN Handling**

  17.3% missingness handled transparently. No row dropping, no pre-imputation. Ideal for materials databases with natural sparsity.

- **Near-Perfect Discrete Imputation**

  E Above Hull R2=0.973, Crystal System R2=0.954, Laue Class R2=0.937. 5,284 missing bulk modulus values imputed from electronic/structural data.

- **Anomaly Detection with Attribution**

  Identifies materials with unusual property combinations. Cell-level attribution pinpoints WHICH properties drive the anomaly.

- **GPU Acceleration**

  10-100x faster than CPU CrossCat via JAX JIT + vmap + pmap. Full analysis of 7,327 materials x 23 columns in ~10 hours on 2xT4.

Key Metrics at a Glance:

Dataset	7,327 materials x 23 properties (Materials Project v2025.09.25)
Column Types	18 continuous, 2 binary, 2 categorical, 1 ordinal
Missingness	17.3% (natural sparsity, no imputation needed)
Inference	10 chains x 500 sweeps on 2xT4 GPUs (~10 hours)
Views Discovered	4 (core properties, symmetry/structure, stability, Poisson ratio)
Imputation R2 (best)	0.973 (E above hull), 0.954 (crystal system), 0.937 (Laue class)
Elasticity Imputed	5,284 bulk modulus + 5,303 shear modulus values
Strongest MI	Laue class <-> crystal system: Linfoot = 0.530
Most Anomalous	Na4S2O5 (typicality = 0.050) -- anomalous dielectric for alkali compound

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Resources

- GitHub: https://github.com/sambhal-labs/jaxcross
- Documentation: https://sambhal-labs.github.io/jaxcross/
- Data Source: https://materialsproject.org/ (API v2025.09.25)
- Full Notebook: examples/materials_project_discovery.ipynb